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Variance

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Average income in the United States

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Background

You have the annual personal income of 11 people from the USA. You have the mean income from the exercise on mean, median and mode

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Task 1

Decide whether you have to use sample or population formula for the variance

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Task 2

Calculate the variance of their income

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Task 3

Generally, what does this number tell you?

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Solution:

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Annual income	Mean	\$ 189,848.18
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12	\$ 62,000.00
13	\$ 64,000.00
14	\$ 49,000.00
15	\$ 324,000.00
16	\$ 1,264,000.00
17	\$ 54,330.00
18	\$ 64,000.00
19	\$ 51,000.00
20	\$ 55,000.00
21	\$ 48,000.00
22	\$ 53,000.00

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Task 1: The question is asking if this is a sample or a population. In other words, are those all the people in the US, receiving salaries?
Obviously not. This is a sample, drawn from the population of all working people in the USA.

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Task 2: Variance \$² 133,433,409,536.36

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Task 3: There is great dispersion between the income of different people in the USA.

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